Geometric Hybrid Dynamical Systems on Intrinsic Manifolds: Part II – Hybrid Invariance Principle

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***Abstract*— This paper establishes a hybrid invariance principle for geometric hybrid dynamical systems, modeled as hybrid inclusions, evolving on intrinsic C^1-manifolds. Building on previous results on Lyapunov stability for hybrid systems presented in the companion paper [1], we introduce the theoretical tools required for invariance-based stability analysis. These include set-convergence concepts for manifolds, characterization of nominal well-posedness, and analysis of ω-limit set properties for precompact solutions. Our variational analysis approach preserves the intrinsic geometric structure of the underlying space while handling set-valued dynamics without requiring uniqueness of solutions. This theory particularly addresses challenges in disconnected manifolds and quotient spaces.**

I. Introduction

Hybrid dynamical systems, which combine continuous-time and discrete-time dynamics, have proven fundamental in modeling phenomena across engineering disciplines [2]–[4]. These systems effectively describe processes involving continuous evolution and abrupt changes, such as mechanical systems with impacts [5], and networked control systems with intermittent communication [6]. The hybrid inclusions framework represents a powerful approach within this field, capturing set-valued dynamics in Euclidean spaces and enabling rigorous stability analysis and control design [2]. Despite these advances in Euclidean settings, many engineering systems inherently evolve on nonlinear spaces, motivating the extension of hybrid dynamical systems to manifolds. Examples include attitude control of rigid bodies on $\mathrm{SO}(3)$ [7], and optimization algorithms on Grassmann manifolds [8].

Previous approaches have addressed hybrid inclusions on manifolds via projections or embeddings [9], considering parallelizable manifolds such as Lie groups [10], [11], or presenting coordinate-dependent implementations [12]. These approaches have proven effective for applications in compact connected manifolds with natural embeddings, such as the 2-sphere $\mathbb{S}^2$ in $\mathbb{R}^3$, the 3-sphere $\mathbb{S}^3$ in $\mathbb{R}^4$, or $\mathrm{SO}(3)$ in $\mathbb{R}^{3\times 3}$.

However, as described in [1], intrinsic approaches that do not rely on coordinates from an ambient space may be preferred when dealing with disconnected manifolds or when studying quotient manifolds. Disconnected manifolds appear in hybrid systems with multiple operational modes on separate manifolds, where intrinsic formulations capture the fundamental topology without artificial connections introduced by embeddings or arbitrary metric choices. Quotient manifolds benefit from intrinsic approaches, as they encode important invariances that embeddings may not preserve. The Grassmann manifold $\mathrm{Gr}(k,n)$ exemplifies this case, with each point representing a k-dimensional linear subspace of the n-dimensional Euclidean space. Though these points could be embedded as projection matrices in $\mathbb{R}^{n\times n}$, such representation obscures the quotient structure $\mathrm{Gr}(k,n) = \mathrm{O}(n)/(\mathrm{O}(k)\times \mathrm{O}(n-k))$, where $O(n)$ denotes the n-dimensional orthogonal group. Indeed, while Whitney's embedding theorem guarantees the *existence* of embeddings in Euclidean space for any smooth manifold [13, Thm. 6.15], it provides no characterization of such embeddings, significantly limiting their practical value in optimization and control applications.

Building on the intrinsic approach for geometric hybrid dynamical systems, this paper complements the work presented in the companion paper [1], which establishes a Lyapunov theorem for such systems. The main contribution of this work is the development of a hybrid invariance principle for intrinsic manifolds. To achieve this, we compile and extend key set-valued analysis concepts and tools from the hybrid inclusions framework in Euclidean space to the intrinsic C^k-manifold setting. Additionally, we characterize nominal well-posedness of hybrid dynamical systems as well as suitable properties of ω-limit sets for solutions that are precompact and defined for arbitrarily large time, providing the necessary groundwork for the invariance principle. Due to space constraints, proofs will be presented elsewhere.

II. Preliminaries

We use $\mathbb{R}$ and $\mathbb{R}_{\geq 0}$ to denote the set of real and nonnegative real numbers, respectively. The set of natural numbers, including 0, is denoted by $\mathbb{N}$. A function $f:\mathbb{R}^m\to\mathbb{R}^n$ is said to be a C^k-function if it has k continuous derivatives. A function $\sigma: X\to\mathbb{R}_{\geq 0}$ is positive definite with respect to $\mathcal{A}\subset X$, denoted by $\sigma\in\mathcal{PD}(\mathcal{A})$, if $\sigma(x)>0$ for all $x\in X\setminus\mathcal{A}$ and $\sigma(x)=0$ for all $x\in\mathcal{A}$. The composition of two maps $f:X\to Y$ and $g:Y\to Z$ is denoted by $g\circ f: X\to Z$. A set-valued map $F:X\rightrightarrows Y$ associates every element in X with a subset of Y. The domain, range, and graph of F are defined as $\operatorname{dom} F := \{x: F(x)\neq\varnothing\}$, $\operatorname{rge} f := \{y:\exists x \text{ s.t. } y\in F(x)\}$, and $\operatorname{gph} F := \{(x,y): y\in F(x)\}$, respectively. Let $(\mathcal{X},\mathcal{T})$ be a topological space. For a set $S\subset\mathcal{X}$, we write $\overline{S}$ for its closure, and $\operatorname{int} S$ for its

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Research partially supported by NSF Grants no. CNS-2039054 and CNS-2111688, by AFOSR Grants nos. FA9550-19-1-0169, FA9550-20-1-0238, FA9550-23-1-0145, and FA9550-23-1-0313, by AFRL Grant nos. FA8651-22-1-0017 and FA8651-23-1-0004, by ARO Grant no. W911NF-20-1-0253, and by DoD Grant no. W911NF-23-1-0158.

interior. A set $S \subset \mathcal{X}$ is *compact* if each open cover of S has a finite subcover, and it is *precompact* if $\overline{S}$ is compact. Given a map $f : \mathcal{X} \to \mathcal{Y}$ and a set $B \subset \mathcal{Y}$, the preimage of B under f is given by $f^{-1}(B) \coloneqq \{x : f(x) \in B\}$. For a function $V : \mathcal{X} \to \mathbb{R}$ and $c \in \mathbb{R}_{\geq 0}$, we let $L_V(c) \coloneqq V^{-1}((-\infty, c])$ denote its c-sublevel set. A map $f : \mathcal{X} \to \mathcal{Y}$ between topological spaces is proper if $f^{-1}(K)$ is compact for every compact set K. Given $x \in \mathcal{X}$, let $\mathrm{Nbd}(x) \coloneqq \{V \in \mathcal{T} : x \in V\}$ be the set of open neighborhoods of x.

An n-dimensional topological manifold $\mathcal{M}$ is a second-countable Hausdorff space that is locally Euclidean of dimension n. A coordinate chart of $\mathcal{M}$ at x is a pair (U, φ) where $x \in U \subset M$, with U an open set and $\varphi : U \to \varphi(U) \subset \mathbb{R}^n$ a homeomorphism. Two charts (U, φ) and (V, ψ) of $\mathcal{M}$ are C^k-compatible if $U \cap V = \varnothing$ or $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is a C^k-diffeomorphism. A C^k-atlas of $\mathcal{M}$ is one whose charts are C^k-compatible. A C^k-manifold $\mathcal{M}$ is a topological manifold endowed with a maximal C^k-atlas. The tangent space to $\mathcal{M}$ at $x \in \mathcal{M}$ is denoted by $\mathrm{T}_x\mathcal{M}$, and the tangent bundle of $\mathcal{M}$ is denoted by $\mathrm{T}\mathcal{M}$. For more details on differential manifolds we refer the reader to [13].

III. Geometric Hybrid Dynamical Systems

Building on Part I [1], we study geometric hybrid systems on manifolds described by the hybrid inclusion

$$\mathcal{H} : \begin{cases} \dot{x} \in F(x) & x \in C \\ x^+ \in G(x) & x \in D \end{cases}, \tag{1}$$

where $\mathcal{M}$ is a C^1-manifold, $x \in \mathcal{M}$ is the state of the system, $C \subset \mathcal{M}$ is the flow set, $D \subset \mathcal{M}$ is the jump set, and $G : D \rightrightarrows \mathcal{M}$ is the jump map. The flow map is a set-valued map satisfying $F(x) \subset \mathrm{T}_x\mathcal{M}$ for each $x \in C$, and is denoted as $F : C \rightrightarrows \mathrm{T}\mathcal{M}$. We use $\mathcal{H} = (C, F, D, G, \mathcal{M})$ to represent a hybrid system using this data.

Solutions to $\mathcal{H}$ are defined on *hybrid time domains*. A set $E \subset \mathbb{R}_{\geq 0} \times \mathbb{N}$ is a hybrid time domain if it is the union of compact hybrid time domains E_j with the property that $E_0 \subset E_1 \subset E_2 \subset \ldots \subset E_j \ldots$. A set $E \subset \mathbb{R}_{\geq 0} \times \mathbb{N}$ is a compact hybrid time domain if there exists $J \in \mathbb{N}$ such that $E = \bigcup_{j=0}^{J} [t_j, t_{j+1}] \times \{j\}$, for some finite sequence of times $\{t_j\}_{j=0}^{J+1}$ satisfying $0 = t_0 \leq t_1 \leq t_2 \leq \ldots \leq t_J \leq t_{J+1}$.

A function $\phi : \mathrm{dom}\,\phi \to \mathcal{M}$ is a hybrid arc if $\mathrm{dom}\,\phi$ is a hybrid time domain and if for each $j \in \mathbb{N}$, the function $t \mapsto \phi(t, j)$ is locally absolutely continuous[1] on the interval $I^j = \{t : (t, j) \in \mathrm{dom}\,\phi\}$. Given a hybrid arc ϕ, its total time of flow is given by $\sup_t \mathrm{dom}\,\phi \coloneqq \sup\{t : (t, j) \in \mathrm{dom}\,\phi\}$. Similarly, the number of times ϕ jumps is given by $\sup_j \mathrm{dom}\,\phi \coloneqq \sup\{j : (t, j) \in \mathrm{dom}\,\phi\}$. Then, the length of its domain is defined as $\mathrm{length}(\mathrm{dom}\,\phi) \coloneqq \sup_t \mathrm{dom}\,\phi + \sup_j \mathrm{dom}\,\phi$.

Definition 1 (Solution to a hybrid system)**.** A hybrid arc ϕ is a solution to the geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ if $\phi(0, 0) \in \overline{C} \cup D$ and the following conditions hold: i) For each $j \in \mathbb{N}$ such that $I^j \coloneqq \{t : (t, j) \in \mathrm{dom}\,\phi\}$ has nonempty interior, ϕ satisfies $\phi(t, j) \in C$ for all $t \in \mathrm{int}\, I^j$ and $\dot{\phi}(t, j) \in F(\phi(t, j))$ for almost all $t \in I^j$; ii) For each $(t, j) \in \mathrm{dom}\,\phi$ such that $(t, j + 1) \in \mathrm{dom}\,\phi$, ϕ satisfies $\phi(t, j) \in D$ and $\phi(t, j + 1) \in G(\phi(t, j))$.

A solution ϕ is *nontrivial* if $\mathrm{dom}\,\phi$ contains at least two points. It is said to be *maximal* if no other solution ψ to $\mathcal{H}$ exists such that $\mathrm{dom}\,\phi$ is a proper subset of $\mathrm{dom}\,\psi$ and $\phi(t, j) = \psi(t, j)$ for all $(t, j) \in \mathrm{dom}\,\phi$. A solution ϕ is said to be *complete* if $\mathrm{length}(\mathrm{dom}\,\phi) = \infty$. The set of solutions (resp., maximal solutions) to $\mathcal{H}$ from a nonempty set $K \subset \mathcal{M}$ is denoted by $\hat{\mathcal{S}}_{\mathcal{H}}(K)$ (resp., $\mathcal{S}_{\mathcal{H}}(K)$).

Now, we define locally Lipschitz Lyapunov function candidates for geometric hybrid systems. The geometric definition of local Lipschitz continuity of a function between C^1-manifolds is presented in [1, Def. 5].

Definition 2 (Lyapunov function candidate)**.** Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a geometric hybrid system. Given nonempty sets $\mathcal{U}, \mathcal{A} \subset \mathcal{M}$, a function $V : \mathrm{dom}\, V \to \mathbb{R}$ is called a *Lyapunov function candidate on $\mathcal{U}$ with respect to $\mathcal{A}$ for $\mathcal{H}$* if the following conditions hold:

1) $(\overline{C} \cup D \cup G(D)) \cap \mathcal{U} \subset \mathrm{dom}\, V$;
2) $\mathcal{U}$ contains an open neighborhood of $\mathcal{A} \cap (\overline{C} \cup D \cup G(D))$;
3) V is continuous on $\mathcal{U}$ and locally Lipschitz on an open set containing $\overline{C} \cup \mathcal{U}$;
4) V is positive definite on $C \cup D \cup G(D)$ with respect to $\mathcal{A}$.

As shown in [1, Prop. 1], locally Lipschitz functions between C^1-manifolds are differentiable almost everywhere. This property enables the definition of generalized directional derivatives [15, Eq. 3.8] to define the rate of change of Lyapunov function candidates in geometric hybrid systems.

Definition 3 ($\dot{V}$ and ΔV)**.** Consider a geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$, sets $\mathcal{U}, \mathcal{A} \subset \mathcal{M}$, and a function $V : \mathrm{dom}\, V \to \mathbb{R}$ defining a Lyapunov function candidate on $\mathcal{U}$ with respect to $\mathcal{A}$ for $\mathcal{H}$.

- The change of V along flows is given by

$$\dot{V}(x) \coloneqq \sup_{v \in F(x) \cap \mathrm{T}_C^{\mathcal{M}}(x)} V^\circ(x, v) \quad \forall x \in C \cap \mathcal{U}, \tag{2}$$

  where $V^\circ(x, v)$ is the generalized directional derivative of V at x in the direction $v \in \mathrm{T}_x\mathcal{M}$, and $\mathrm{T}_C^{\mathcal{M}}(x)$ denotes the tangent cone to C at x [1, Def. 4].
- The change of V at jumps is given by

$$\Delta V(x) \coloneqq \sup_{g \in G(x)} V(g) - V(x) \quad \forall x \in D \cap \mathcal{U}. \tag{3}$$

These definitions were central to the hybrid Lyapunov theorem for geometric hybrid systems [1, Thm. 1], which we reproduce below.

Theorem 1. *(Hybrid Lyapunov theorem). Let $\mathcal{A} \subset \mathcal{M}$ be a nonempty compact set, and $V : \mathrm{dom}\, V \to \mathbb{R}_{\geq 0}$ be a Lyapunov function candidate with respect to $\mathcal{A}$ for a geometric hybrid dynamical system $\mathcal{H} = (C, F, D, G, \mathcal{M})$. The set $\mathcal{A}$*

[1] A function $\gamma : \mathbb{R} \supset I \to \mathcal{M}$ is said to be locally absolutely continuous if, for each C^1-smooth function $\psi : \mathcal{M} \to \mathbb{R}$, the composition $\psi \circ \gamma : I \to \mathbb{R}$ is locally absolutely continuous [14, Section A.2.1].

is uniformly globally pre-asymptotically stable for $\mathcal{H}$ if the Lyapunov function candidate V is a proper map, and

$$\dot{V}(x) \leq -\rho_C(x) \quad \forall x \in C,$$
$$\Delta V(x) \leq -\rho_D(x) \quad \forall x \in D,$$

for some lower semicontinuous functions $\rho_C, \rho_D \in \mathcal{PD}(\mathcal{A})$.

In practice, the decrease conditions of Theorem 1 may only be satisfied with $\dot{V} \leq 0$ and $\Delta V \leq 0$. The hybrid invariance principle [2, Ch. 8] resolves this limitation when $\mathcal{M} = \mathbb{R}^n$, but no comparable result exists for hybrid systems on manifolds. This paper presents the invariance principle on C^1-manifolds for geometric hybrid systems modeled via hybrid inclusions of the form in (1). To this end, we first introduce some tools from set-valued analysis.

IV. Set-valued Analysis on Manifolds

Set-valued analysis has been foundational for analyzing the behavior of solutions in the hybrid inclusions framework in Euclidean spaces [2, Ch. 5, 6]. This section presents several tools for set-valued analysis on topological manifolds. We start with a notion of set convergence that reduces to the limits of sequences of sets from [2, Def. 5.1] when $\mathcal{M} = \mathbb{R}^n$.

Definition 4 (Set convergence [16, p.209])**.** Let $\mathcal{M}$ be a topological manifold and $\{S_n\}_{n=1}^{\infty}$ a sequence of sets in $\mathcal{M}$.

- The *inner limit* of the sequence $\{S_n\}_{n=1}^{\infty}$, denoted by $\liminf_{n\to\infty} S_n$, is defined as follows:

$$\liminf_{n\to\infty} S_n \coloneqq \left\{ x \in \mathcal{M} : \begin{array}{l} \forall V \in \mathrm{Nbd}(x), \exists i_0 > 0 \\ \text{s.t. } V \cap S_i \neq \varnothing \ \ \forall i > i_0 \end{array} \right\}.$$

- The *outer limit* of the sequence $\{S_n\}_{n=1}^{\infty}$, denoted by $\limsup_{n\to\infty} S_n$, is defined as follows:

$$\limsup_{n\to\infty} S_n \coloneqq \left\{ x \in \mathcal{M} : \begin{array}{l} \forall V \in \mathrm{Nbd}(x), \forall i_0 > 0, \\ \exists i > i_0 \text{ s.t. } V \cap S_i \neq \varnothing \end{array} \right\}.$$

The sequence $\{S_n\}_{n=1}^{\infty}$ is said to converge if its inner limit and outer limit are equal, and the limit is given by $\lim_{n\to\infty} S_n = \liminf_{n\to\infty} S_n = \limsup_{n\to\infty} S_n$.

The following definition of graphical convergence is taken verbatim from [2, Def. 5.18] and generalizes to C^1-manifolds.

Definition 5 (Graphical convergence)**.** Given topological manifolds $\mathcal{M}$ and $\mathcal{N}$, we say that a sequence $\{M_i\}_{i=1}^{\infty}$ of set-valued mappings $M_i : \mathcal{M} \rightrightarrows \mathcal{N}$ converges *graphically* if the sequence of sets $\{\mathrm{gph}(M_i)\}_{i=1}^{\infty}$ converges in $\mathcal{M} \times \mathcal{N}$ in the sense of Definition 4. The graphical limit $\text{gph-lim}_{i\to\infty} M_i$ of a graphically convergent sequence $\{M_i\}_{i=1}^{\infty}$ is the mapping $M : \mathcal{M} \rightrightarrows \mathcal{N}$ such that $\mathrm{gph}\, M = \lim_{i\to\infty} \mathrm{gph}\, M_i$.

By extrapolating the notion for the Euclidean case [17, Cor. 4.11], we say that the sequence $\{S_n\}_{n=1}^{\infty}$ of nonempty sets in $\mathcal{M}$ *escapes to the horizon* if for each $K \subset \mathcal{M}$ compact, there exists $n_0 > 0$ such that $S_n \cap K = \varnothing$ for each $n \geq n_0$.

Lemma 1. *Given a topological manifold $\mathcal{M}$, every sequence $\{S_i\}_{i=1}^{\infty}$ of nonempty subsets of $\mathcal{M}$ either escapes to the horizon or has a subsequence converging to a nonempty set.*

The following definition considers sequences of maps whose graphs do not escape to the horizon.

Definition 6 (Local eventual precompactness)**.** A sequence $\{M_i\}_{i=1}^{\infty}$ of set-valued maps $M_i : \mathcal{M} \rightrightarrows \mathcal{N}$ between topological manifolds is *locally eventually precompact* if, for each compact set $K \subset \mathcal{M}$, there exist $i_0 > 0$ and a compact set $K' \subset \mathcal{N}$ such that $M_i(K) \subset K'$ for each $i > i_0$.

The following lemma, which is similar to [2, Example 5.19], will be instrumental when considering the convergence of sequences of solutions to hybrid systems.

Lemma 2. *Let $\mathcal{M}$ and $\mathcal{N}$ be topological manifolds. Consider a sequence $\{M_i\}_{i=1}^{\infty}$ of set-valued maps $M_i : \mathcal{M} \rightrightarrows \mathcal{N}$ that is graphically convergent, and let $M = \text{gph-lim}_{i\to\infty} M_i$. Then,*

$$\mathrm{dom}\, M \subset \lim_{i\to\infty} \mathrm{dom}\, M_i \quad \textit{and} \quad \mathrm{rge}\, M \subset \lim_{i\to\infty} \mathrm{rge}\, M_i. \tag{4}$$

If, in addition, the sequence $\{M_i\}_{i=1}^{\infty}$ is locally eventually precompact, then $\mathrm{dom}\, M = \lim_{i\to\infty} \mathrm{dom}\, M_i$.

These notions of graphical convergence on manifolds enable us to introduce nominal well-posedness for geometric hybrid systems, as defined in the next section.

V. Nominally Well-Posed Geometric Hybrid Dynamical Systems

Properties of ω-limit sets of solutions to nominal well-posed hybrid systems are crucial for proving the hybrid invariance principle in Euclidean settings [2, Ch. 6]. First, we introduce truncation and graphical convergence of hybrid arcs before formally defining nominally well-posed geometric hybrid systems on C^1-manifolds.

Definition 7 (Truncation of a hybrid arc)**.** Given $\tau > 0$ and a hybrid arc $\phi : \mathrm{dom}\, \phi \to \mathcal{M}$, the truncation of ϕ up to τ is the hybrid arc $\phi|_{\leq\tau} : \mathrm{dom}\, \phi|_{\leq\tau} \to \mathcal{M}$ with $\mathrm{dom}\, \phi|_{\leq\tau} \coloneqq \{(t,j) \in \mathrm{dom}\, \phi : t + j \leq \tau\}$ and $\phi|_{\leq\tau}(t,j) = \phi(t,j)$ for all $(t,j) \in \mathrm{dom}\, \phi|_{\leq\tau}$. The definitions of $\phi|_{<\tau}$, $\phi|_{\geq\tau}$, and $\phi|_{>\tau}$ are given similarly by replacing "$\leq$" with "$<$", "$\geq$", and "$>$", respectively.

Graphical convergence of a sequence $\{\phi_i\}_{i=1}^{\infty}$ of hybrid arcs $\phi_i : \mathrm{dom}\, \phi_i \to \mathcal{M}$ is defined according to Definition 5, and the graphical limit is the set-valued map $\phi : \mathbb{R} \times \mathbb{N} \rightrightarrows \mathcal{M}$ denoted by $\phi = \text{gph-lim}_{i\to\infty} \phi_i$. The following lemma provides conditions under which such convergence is guaranteed: every sequence of hybrid arcs that does not escape to the horizon has a convergent subsequence. An equivalent result for hybrid arcs on the Euclidean space is presented in [2, Theorem 6.1].

Lemma 3. *For each sequence of hybrid arcs $\{\phi_i\}_{i=1}^{\infty}$, where $\phi_i : \mathrm{dom}\, \phi_i \to \mathcal{M}$, such that there exists a compact set $K \subset \mathcal{M}$ satisfying $\phi_i(0,0) \in K$ for each $i \in \mathbb{N}$, there exists a subsequence of $\{\phi_i\}_{i=1}^{\infty}$ that converges graphically.*

Now, we introduce the concept of nominal well-posedness, extending the notion in [2, Definition 6.2] to C^1-manifolds. This property ensures that the graphical limit of a sequence of solutions to $\mathcal{H}$ is also a solution to $\mathcal{H}$.

Definition 8 (Nominal well-posedness of $\mathcal{H}$)**.** A geometric hybrid system $\mathcal{H}$ is said to be *nominally well-posed* if the following holds: for each graphically convergent sequence $\{\phi_i\}_{i=1}^{\infty}$ of solutions $\phi_i : \operatorname{dom}\phi_i \to \mathcal{M}$ to $\mathcal{H}$ with $\lim_{i\to\infty}\phi_i(0,0) = \xi \in \mathcal{M}$,

(a) if the sequence $\{\phi_i\}_{i=1}^{\infty}$ is locally eventually precompact,[2] then there exists $\tau \in \mathbb{R}_{\geq 0} \cup \{\infty\}$ such that $\lim_{i\to\infty}\operatorname{length}(\operatorname{dom}\phi_i) = \tau$, and $\phi := \operatorname{gph-lim}_{i\to\infty}\phi_i$ is such that $\phi \in \hat{\mathcal{S}}_{\mathcal{H}}(\xi)$ and $\operatorname{length}(\operatorname{dom}\phi) = \tau$;

(b) if the sequence $\{\phi_i\}_{i=1}^{\infty}$ is not locally eventually precompact, then there exists $\tau > 0$ and a sequence $(t_i, j_i) \in \operatorname{dom}\phi_i$, $i = 1, 2, \ldots$, with $\lim_{i\to\infty} t_i + j_i = \tau$ such that

i) for each compact set $K \subset \mathcal{M}$, there exists $i_0 > 0$ such that $\phi_i(t_i, j_i) \notin K$ for each $i > i_0$, and

ii) the hybrid arc $\phi := (\operatorname{gph-lim}_{i\to\infty}\phi_i)|_{<\tau}$ is such that $\phi \in \mathcal{S}_{\mathcal{H}}(\xi)$, $\operatorname{length}(\operatorname{dom}\phi) = \tau$, and for each compact set $K \subset \mathcal{M}$, there exists $\tau' \in (0, \tau)$ such that $K \cap \operatorname{rge}\phi|_{\geq\tau'} = \varnothing$.

VI. Sufficient Conditions for Nominal Well-posedness

Directly verifying nominal well-posedness of a geometric hybrid system $\mathcal{H}$ using Definition 8 is nontrivial. Thus, we provide sufficient conditions to ensure this property, beginning with locally precompact and outer semicontinuous set-valued maps on topological manifolds.

Definition 9 (Local precompactness)**.** Given two topological manifolds $\mathcal{M}$ and $\mathcal{N}$, a set-valued map $S : \mathcal{M} \rightrightarrows \mathcal{N}$ is locally precompact at $x \in \mathcal{M}$ if there exists $V \in \operatorname{Nbd}(x)$ such that $S(V)$ is precompact. The set-valued map S is locally precompact relative to a set $E \subset \mathcal{M}$ if S, restricted to E, is locally precompact at each $x \in E$.

Remark 1. Definition 9 is an extension of the notion of locally bounded set-valued maps [2], [17] to the geometric setting. The terminological change from "boundedness" to "precompactness" ensures consistency when $\mathcal{N}$ therein is a Riemannian manifold that is not metrically complete. In particular, a set may be bounded in such a space, but it may not be precompact; see Hopf-Rinow theorem [18, Thm. 6.19].

Definition 10 (Outer semicontinuity)**.** Given two topological manifolds $\mathcal{M}$ and $\mathcal{N}$, a set-valued map $S : \mathcal{M} \rightrightarrows \mathcal{N}$ is outer semicontinuous at $x \in \mathcal{M}$ if $\limsup_{\bar{x}\to x} F(\bar{x}) \subset F(x)$. The map S is outer semicontinuous relative to a set $E \subset \mathcal{M}$ if S, restricted to E, is outer semicontinuous at each $x \in E$.

Now, we introduce the geometric hybrid basic conditions. For similar conditions in Euclidean spaces, see [2, Thm. 6.8].

Definition 11 (Geometric hybrid basic conditions)**.** A geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ is said to satisfy the geometric hybrid basic conditions if

1) $\mathcal{M}$ is a C^1-manifold;
2) $C \subset \mathcal{M}$ and $D \subset \mathcal{M}$ are closed relative to $\mathcal{M}$;
3) $F : C \rightrightarrows \mathrm{T}\mathcal{M}$ is outer semicontinuous and locally precompact relative to C, and $F(x)$ is convex for all $x \in C$;
4) $G : D \rightrightarrows \mathcal{M}$ is outer semicontinuous and locally precompact relative to D.

We show that the geometric hybrid basic conditions are sufficient to ensure nominal well-posedness of $\mathcal{H}$ when the sequence of solutions to $\mathcal{H}$ is locally eventually precompact.

Theorem 2. *Let the geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ satisfy the geometric hybrid basic conditions in Definition 11. Consider a graphically convergent and locally eventually precompact sequence $\{\phi_i\}_{i=1}^{\infty}$ of solutions $\phi_i : \operatorname{dom}\phi_i \to \mathcal{M}$ to $\mathcal{H}$. Then, the graphical limit $\phi = \operatorname{gph-lim}_{i\to\infty}\phi_i$ is a solution to $\mathcal{H}$.*

VII. Invariance Principle for Geometric Hybrid Dynamical Systems

In this section, we present the hybrid invariance principle for geometric hybrid systems. First, we establish properties of ω-limit sets of complete and precompact[3] solutions to nominally well-posed hybrid systems. For an equivalent characterization when $\mathcal{M} = \mathbb{R}^n$, see [2, Ch.6], [19].

Definition 12 (Weak invariance)**.** Given a geometric hybrid system $\mathcal{H}$, a nonempty set $K \subset \mathcal{M}$ is said to be

1) *weakly forward invariant* for $\mathcal{H}$ if, for each $\xi \in K$, there exists a complete solution $\phi \in \mathcal{S}_{\mathcal{H}}(\xi)$ with $\operatorname{rge}\phi \subset K$;
2) *weakly backward invariant* for $\mathcal{H}$ if, for each $\xi \in K$ and each $T > 0$, there exists $\phi \in \mathcal{S}_{\mathcal{H}}(K)$ such that, for some hybrid time $(t^\star, j^\star) \in \operatorname{dom}\phi$ with $t^\star + j^\star \geq T$, ϕ satisfies $\phi(t^\star, j^\star) = \xi$ and $\operatorname{rge}\phi|_{\leq t^\star + j^\star} \subset K$;
3) *weakly invariant* for $\mathcal{H}$ if it is both weakly forward invariant and weakly backward invariant.

Definition 13 (ω-limit set of a hybrid arc)**.** The ω-limit set of a hybrid arc $\phi : \operatorname{dom}\phi \to \mathcal{M}$, denoted by $\Omega(\phi)$, is the set of all points $x \in \mathcal{M}$ for which there exists a sequence $\{(t_i, j_i)\}_{i=1}^{\infty} \subset \operatorname{dom}\phi$ with $\lim_{i\to\infty} t_i + j_i = \infty$ such that $\lim_{i\to\infty}\phi(t_i, j_i) = x$.

The following result extends [2, Proposition 6.21] to hybrid systems evolving on C^1-manifolds.

Proposition 1. *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system, and ϕ be a complete and precompact solution to $\mathcal{H}$. Then, $\Omega(\phi)$ is nonempty, compact, weakly invariant, and for each open neighborhood $\mathcal{U}$ of $\Omega(\phi)$, there exists $N > 0$ such that $\phi(t, j) \in \mathcal{U}$ for all $(t, j) \in \operatorname{dom}\phi$ such that $t + j \geq N$.*

Now, given a function $V : \operatorname{dom} V \to \mathbb{R}$ satisfying items 1 and 3 of Definition 2, we define $\dot{V}^{-1}(0) := \{x \in C : \dot{V}(x) = 0\}$, $\Delta V^{-1}(0) := \{x \in D : \Delta V(x) = 0\}$, and for each $r \in \mathbb{R}$, $V^{-1}(r) := \{x \in \mathcal{M} : V(x) = r\}$. With these definitions, we are now prepared to present the main result of the paper.

Theorem 3 (Hybrid invariance principle)**.** *Consider a nominally well-posed geometric hybrid system $\mathcal{H} =$*

[2]Local eventual precompactness of a sequence of hybrid arcs $\{\phi_i\}_{i=1}^{\infty}$ is defined by replacing $\mathcal{M}$ by $\mathbb{R}_{\geq 0} \times \mathbb{N}$, and $\mathcal{N}$ by $\mathcal{M}$ in Definition 6.

[3]A solution ϕ to $\mathcal{H}$ is *precompact* if $\operatorname{rge}\phi \subset \mathcal{M}$ is precompact.

$(C, F, D, G, \mathcal{M})$, a nonempty set $\mathcal{U} \subset \mathcal{M}$, and a function $V : \operatorname{dom} V \to \mathbb{R}$ satisfying items 1 and 3 of Def. 2 such that:

$$\begin{aligned} \dot{V}(x) &\leq 0 \qquad \forall x \in C \cap \mathcal{U}, \\ \Delta V(x) &\leq 0 \qquad \forall x \in D \cap \mathcal{U}. \end{aligned} \tag{5}$$

Let $\phi^\star$ be a complete and precompact solution to $\mathcal{H}$ with $\overline{\operatorname{rge} \phi^\star} \subset \mathcal{U}$. Then, for some $r \in V(\mathcal{U} \cap (C \cup D \cup G(D)))$, the set $\Omega(\phi^\star)$ is the largest weakly invariant set in

$$V^{-1}(r) \cap \mathcal{U} \cap \Big[\dot{V}^{-1}(0) \cup \big(\Delta V^{-1}(0) \cap G(\Delta V^{-1}(0))\big)\Big]. \tag{6}$$

VIII. Illustrative Example

We study the problem of point stabilization on the tangent bundle of the 1-sphere $\mathbb{S}^1$, defined as the quotient space $\mathbb{S}^1 := [0,1]/\{0,1\}$, where the notation "/" indicates that the points $0, 1 \in [0,1] \subset \mathbb{R}$ are taken to be equivalent ($0 \sim 1$). This intrinsic definition of $\mathbb{S}^1$ does not require an embedding in $\mathbb{R}^2$. We highlight this distinction in Figure 1. Furthermore, we endow $\mathbb{S}^1$ with a Riemannian metric and the associated Levi-Civita connection. The tangent bundle to $\mathbb{S}^1$ is denoted by the smooth manifold $\mathcal{M} := \mathrm{T}\mathbb{S}^1 = \mathrm{T}([0,1] \times \mathbb{R})/\sim$, where the equivalence relation $\sim$ on $\mathrm{T}\mathbb{S}^1$ is defined so that $(0, v) \sim (1, v)$ for all $v \in \mathbb{R}$. For more details on quotient spaces, we refer the reader to [16, Ch. 9].

Now, consider a wall on $\mathbb{S}^1$ with non-zero thickness defined by $\mathcal{W} := ([0,\alpha] \cup [1-\alpha, 1])/\{0,1\} \subset \mathbb{S}^1$ for $0 < \alpha < 1/2$ as shown in Figure 1. While the regions $[0,\alpha]$ and $[1-\alpha, 1]$ are disconnected, the identification $0 \sim 1$ makes $\mathcal{W}$ topologically connected on $\mathbb{S}^1$. The problem setup is as follows. A point mass moves on $\mathbb{S}^1$, with its position given by $[z_1] \in \mathbb{S}^1$ and velocity by $[z_2] \in \mathrm{T}_{[z_1]}\mathbb{S}^1$, causing its state $x := ([z_1], [z_2])$ to evolve on the tangent bundle $\mathrm{T}\mathbb{S}^1$. Due to the wall, the position $[z_1]$ of the point mass evolves only on $\overline{\mathbb{S}^1 \setminus \mathcal{W}} = \{[z_1] : z_1 \in [\alpha, 1-\alpha]\}$. The impacts between the point mass and the wall occur when $[z_1] \in \partial\mathcal{W}$ and $[z_2] \in \mathrm{T}^{\mathbb{S}^1}_{\mathcal{W}}([z_1])$, where $\mathrm{T}^{\mathbb{S}^1}_{\mathcal{W}}([z_1])$ denotes the tangent cone to $\mathcal{W}$ at $[z_1] \in \mathbb{S}^1$; see [1, Def. 3.2].

Let $U : \mathbb{S}^1 \to \mathbb{R}_{\geq 0}$ be a potential field on $\mathbb{S}^1$ such that $U \in \mathcal{PD}([0])$. The point mass follows negative gradient of U during flows, while impacts scale the velocity by $\lambda \in [0,1]$ at jumps, giving the hybrid dynamics

$$\widetilde{\mathcal{H}} : \begin{cases} \dot{x} = F(x) & x \in \tilde{C} \\ x^+ \in G(x) := \{([z_1], [-\lambda z_2]) : \lambda \in [0,1]\} & x \in \tilde{D} \end{cases}$$

where $\tilde{C} := \{x \in \mathcal{M} : [z_1] \in \overline{\mathbb{S}^1 \setminus \mathcal{W}}, [z_2] \in \mathrm{T}_{[z_1]}\mathbb{S}^1\}$, $\tilde{D} := \{x \in \mathcal{M} : [z_1] \in \partial\mathcal{W}, [z_2] \in \mathrm{T}^{\mathbb{S}^1}_{\mathcal{W}}([z_1])\}$, and

$$F(x) := \mathfrak{S}(x) + \mathrm{vlft}_x(-\nabla U([z_1]) - b[z_2]) \qquad \forall x \in \tilde{C}.$$

The map $\mathfrak{S} : \mathcal{M} \to \mathrm{T}\mathcal{M}$ denotes the "geodesic spray;" i.e., the unique vector field on $\mathcal{M} = \mathrm{T}\mathbb{S}^1$ whose integral curves, when projected onto $\mathbb{S}^1$ using the canonical tangent bundle projection map $\pi : \mathrm{T}\mathbb{S}^1 \to \mathbb{S}^1$, are geodesics under the Levi-Civita connection; see [14, Remark 3.103]. Furthermore, for each $x \in \mathrm{T}\mathbb{S}^1$, the map $\mathrm{vlft}_x : \mathrm{T}_{\pi(x)}\mathbb{S}^1 \to \mathrm{T}\mathrm{T}_{\pi(x)}\mathbb{S}^1$ denotes the vertical lift[4] at x. Letting Γ^1_{11} denote the Christoffel symbol of the Levi-Civita connection ∇ in $\mathbb{S}^1$, the coordinate expressions for $\mathfrak{S}$ and vlft around $x = ([z_1], [z_2]) \in \mathcal{M}$ are given by

$$\mathfrak{S}(x) = z_2 \frac{\partial}{\partial z_1}\bigg|_x - \Gamma^1_{11} z_2^2 \frac{\partial}{\partial z_2}\bigg|_x, \quad \mathrm{vlft}_x(w) = w_1 \frac{\partial}{\partial z_2}\bigg|_x \tag{7}$$

where z_1, z_2, and w_1 denote, under some chart, the coordinate representations of $[z_1] \in \mathbb{S}^1$, $[z_2] \in \mathrm{T}_{[z_1]}\mathbb{S}^1$, and a tangent vector $w \in \mathrm{T}_{[z_1]}\mathbb{S}^1$ respectively. Note that we have used the fact that $\dim(\mathbb{S}^1) = 1$. We recall from [13, Ch. 3] that, for each $i \in \{1,2\}$ and each $x \in \mathcal{M}$, the operator $\partial/\partial z_i|_x$ constitutes an element of $\mathrm{T}_x\mathcal{M}$.

The constant $b > 0$ characterizes the effect of dissipative forces.[5] This definition of the flow map F follows the presentation in [14, §4.6.3], where the covariant derivative representation of the geometric dynamics of a mechanical system are "lifted" to obtain a first-order ODE on the tangent bundle that is compatible with the flows of (1). As the coordinate representations and the Christoffel symbols depend continuously on x, the flow map F is continuous. The set-valued jump map G accounts for uncertainty in the value of λ.

Note that, while the geometric hybrid systems framework presented in this paper removes dependence from Riemannian metrics, it does not preclude using one.

Pendulum with impacts: We consider the following form for the potential field U:

$$U([z_1]) := \frac{a}{2\pi}(1 - \cos(2\pi z_1)) \qquad \forall z_1 \in [0,1], \tag{8}$$

where $a > 0$. Then, the hybrid system $\widetilde{\mathcal{H}}$ describes the dynamics of a point-mass pendulum rotating in a circle, with impacts occurring at the wall $\mathcal{W}$.

The total energy of the pendulum is given by $\tilde{V}(x) := U([z_1]) + \frac{1}{2} z_2^2$ for all $x \in \mathrm{T}\mathbb{S}^1$, where the term z_2^2 is obtained by using the Riemannian metric. Pick any $r \in (0, \tilde{V}(x'))$, where $x' := ([\frac{1}{2}], [0]) \in \mathrm{T}\mathbb{S}^1$ is the unstable equilibrium point of the continuous-time dynamics of $\mathcal{H}$; i.e., where the pendulum points "up" and its velocity is zero. We assume that each solution to $\widetilde{\mathcal{H}}$ is initialized in the r-sublevel set of $\tilde{V}$, denoted by $L_{\tilde{V}}(r) \subset \mathrm{T}\mathbb{S}^1$. Consider the hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$, where $C = \tilde{C} \cap L_{\tilde{V}}(r)$, $D = \tilde{D} \cap L_{\tilde{V}}(r)$. Using (7), in normal coordinates [18, Prop. 5.24] where $\Gamma^1_{11} = 0$, we have that

$$\begin{aligned} \dot{\tilde{V}}(x) &= -b z_2^2 \leq 0 && \forall x \in C, \\ \Delta \tilde{V}(x) &= \sup\nolimits_{\lambda \in [0,1]} -\tfrac{1-\lambda}{2} z_2^2 \ = 0 && \forall x \in D. \end{aligned} \tag{9}$$

Since $\tilde{V}$ is nonincreasing on a neighborhood of $L_{\tilde{V}}(r)$, each solution that starts in $L_{\tilde{V}}(r)$ stays in it. Also, using [1, Prop. 1], each maximal solution to $\mathcal{H}$ is complete. Then, since $L_{\tilde{V}}(r)$ is compact, the sets C and D are compact, and each $\phi \in \mathcal{S}_{\mathcal{H}}(L_{\tilde{V}}(r))$ is complete and precompact.

For the hybrid system $\mathcal{H}$, we certify pre-asymptotic stability of the compact set $\mathcal{A} := \{([z_1], [z_2]) : [z_1] \in \partial\mathcal{W}, [z_2] = 0\} \subset \mathrm{T}\mathbb{S}^1$, where the pendulum rests at the wall boundary with zero velocity. Let $\mathcal{U} := C \cup D \cup G(D)$, and consider

[4] See [14, Def. 3.73]. Specifically, given $x = (p, v) \in \mathcal{M} = \mathrm{T}\mathbb{S}^1$, define $\mathrm{vlft}_x(w) = \frac{d}{dt}(v + tw)|_{t=0}$ for each $w \in \mathrm{T}_p\mathbb{S}^1$.

[5] A dissipative force [14, Def. 4.65] is obtained by mapping $-b[z_2] \in \mathrm{T}\mathbb{S}^1$ to the cotangent bundle $\mathrm{T}^*\mathbb{S}^1$ by the *flat musical isomorphism* [14, §2.3.4].

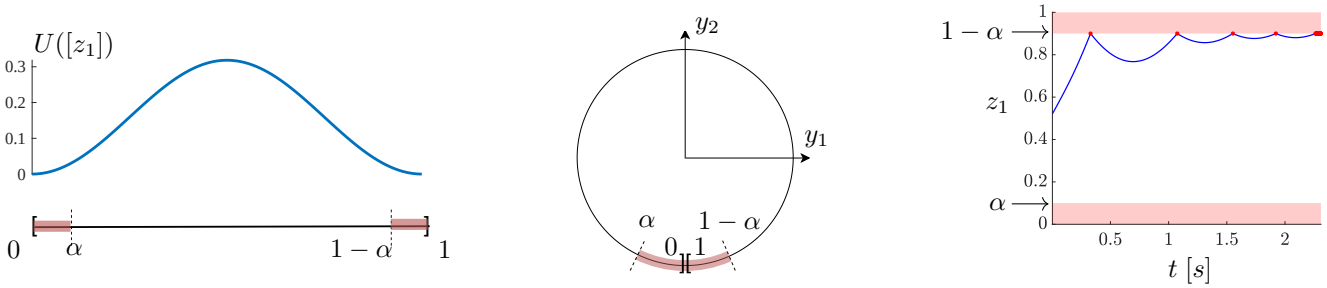


Fig. 1: *Left:* Illustration of the intrinsic 1-sphere $\mathbb{S}^1 = [0,1]/\{0,1\}$. The numbers below the $[0,1]$ segment represent values of $z_1 \in [0,1]$. The red region denotes the wall $\mathcal{W}$. The plot on top denotes the potential field U in (8). *Center:* Illustration of the unit circle $\mathbb{S}^1$ as embedded in $\mathbb{R}^2$, i.e., $\{(y_1, y_2) \in \mathbb{R}^2 : y_1^2 + y_2^2 = 1\}$. The numbers therein represent values of $z_1 \in [0,1]$. By the identification of $z_1 = 0$ and $z_1 = 1$, they represent the same point. *Right:* Plot of z_1 vs. time for a sample solution. The solution converges to the boundary of the wall $\mathcal{W}$.

the Lyapunov function candidate $V : \mathrm{T}\mathbb{S}^1 \to \mathbb{R}_{\geq 0}$ on $\mathcal{U}$ with respect to $\mathcal{A}$ for $\mathcal{H}$, defined by $V(x) := \tilde{V}(x) - \tilde{V}(\mathcal{A})$ for all $x \in \mathrm{T}\mathbb{S}^1$. Note that $V(x) < 0$ only if $x \in \operatorname{int} \mathcal{W}$. Due to the definitions of the sets C and D, $\operatorname{int} \mathcal{W}$ is not contained in $C \cup D \cup G(D)$. Thus, V is positive definite with respect to $\mathcal{A}$ on $C \cup D \cup G(D)$, and from (9), V satisfies (5).

With C and D closed in $\mathrm{T}\mathbb{S}^1$, F single-valued and continuous, and G locally precompact and outer semicontinuous on D (since $\lambda \in [0,1]$), $\mathcal{H}$ satisfies Definition 11. Additionally, since $L_{\tilde{V}}(r)$ is compact and forward invariant for $\mathcal{H}$ as it is forward invariant for $\widetilde{\mathcal{H}}$, each sequence of solutions to $\mathcal{H}$ is locally eventually precompact. Then, $\mathcal{H}$ is nominally well-posed due to Theorem 2. As $G(D) \subset C \cup D$, we also have that each solution ϕ to $\mathcal{H}$ satisfies $\overline{\operatorname{rge} \phi} \subset \mathcal{U}$. Pick any solution ϕ to $\mathcal{H}$. Since $\dot{V}^{-1}(0) = \{x \in C : [z_2] = 0\}$, $\Delta V^{-1}(0) = \{x \in D : [z_2] = 0\} = \mathcal{A}$, and $G(\Delta V^{-1}(0)) = \mathcal{A}$, the largest weakly invariant set in (6) is the set $\mathcal{A}$, as, for each $\xi \in \mathcal{A}$, each $\phi \in \mathcal{S}_{\mathcal{H}}(\xi)$ satisfies $\operatorname{rge} \phi = \{\xi\}$. Then, by Theorem 3, ϕ converges to $\mathcal{A}$. Since the choice of ϕ is arbitrary and $\mathcal{U} = C \cup D \cup G(D)$, $\mathcal{A}$ is globally attractive for $\mathcal{H}$. Also, since V is proper and nonincreasing along solutions, $\mathcal{A}$ is stable for $\mathcal{H}$. Then, $\mathcal{A}$ is globally asymptotically stable for $\mathcal{H}$; see Figure 1.

IX. Conclusion

We augment the toolset developed in Part I [1] by leveraging the properties of ω-limit sets of complete and precompact solutions to present a hybrid invariance principle for geometric hybrid dynamical systems on intrinsic manifolds. In doing so, we employ set-valued analysis to present the notion of nominally well-posed geometric hybrid dynamical systems. Sufficient conditions for nominal well-posedness, defined by locally eventually precompact sequences of solutions, are discussed. Future work will study well-posedness when such sequences are not locally eventually precompact and for perturbed geometric hybrid dynamical systems.

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